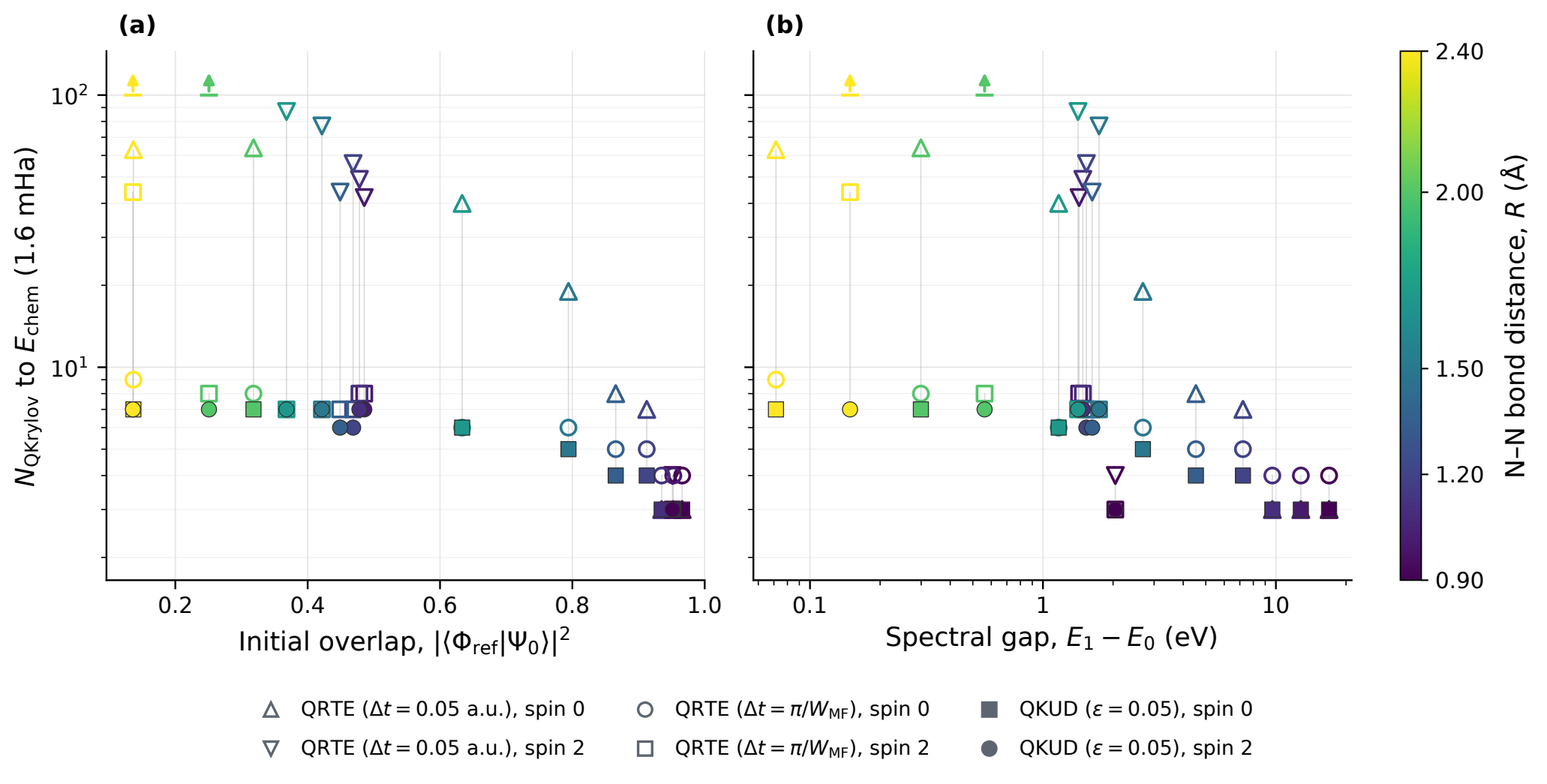

(a)
(b)
N_{QKrylov} to E_{chem} (1.6 mHa)
10^2
10^1
0.2
0.4
0.6
0.8
1.0
Initial overlap, $|\langle\Phi_{\mathrm{ref}}|\Psi_0\rangle|^2$
0.1
1
10
Spectral gap, $E_1 - E_0$ (eV)
N–N bond distance, R (Å)
2.40
2.00
1.50
1.20
0.90
QRTE ($\Delta t = 0.05$ a.u.), spin 0
QRTE ($\Delta t = \pi/W_{\mathrm{MF}}$), spin 0
QKUD ($\varepsilon = 0.05$), spin 0
QRTE ($\Delta t = 0.05$ a.u.), spin 2
QRTE ($\Delta t = \pi/W_{\mathrm{MF}}$), spin 2
QKUD ($\varepsilon = 0.05$), spin 2